Large Scale Genotype by Environment Model Development Competition Explores New Ways of Predicting Maize Yield from Genomic Data

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Abstract

Farm productivity and profitability require varieties with maximum yield, but measuring yield at scale is costly, especially where each cultivar must be trialed in many environments to predict future performance. Genomic yield prediction is a critical method for identifying and investing phenotyping resources in the most promising cultivars. Using an unprecedented public data set containing ten years of public maize yield data, across 272 environments, a large-scale yield prediction competition was designed and completed around the Genomes to Fields (G2F) project. Around 370 participants competed to develop new and innovative genomic and environomic prediction models to predict grain yield for 959 new genotypes in 23 new environments. Participant models used a large variety of methods and approaches, of which a

Multivariate Mixed Linear Model was the most predictive in new trials. The winner and other competition participants share here their successful models with the plant breeding community. Post-competition ensemble models demonstrate the complementary properties of different models in the competition, further improved results by XX%. Given the unprecedented amount of data involved in improving predictions, the top [models should be used in tandem/ ensemble for future work across crops] OR [model should be tested in future maize studies].

Introduction

Genomic prediction/selection (GP/GS) uses genome-wide markers to predict the breeding values or performance of individuals, enabling breeders to accelerate genetic gain in crop improvement (Meuwissen et al., 2001; Crossa et al., 2018). Since its initial conception, GP/GS has been implemented in crop breeding offering a faster and potentially more accurate alternative to conventional phenotypic selection by shortening breeding cycles and reducing phenotyping costs (Crossa et al., 2021). This has been further validated through rapid industry adoption of the GP/GS approach, but is rarely published due to competitive concerns (REF OR Pers. Com.). The adoption of GS in crops has demonstrated increased selection efficiency and genetic gain compared with traditional phenotypic selection methods (Bandeira e Sousa et al., 2017, Jines et al., 2025). However, realizing GP/GS gains in practice requires models to perform robustly across the wide range of environments where that crop is grown as well as predict future untested environments and genotypes. Genotype-by-environment (G×E) interactions, where the relative performance of genotypes changes across environments, are a major challenge for both plant breeding programs and GP/GS models (Rogers & Holland, 2022). GxE interactions force breeders to conduct costly multi-environment trials to identify superior, stable cultivars for target regions (Acosta-Pech et al., 2017). Developing genomic prediction models that effectively combine information across environments in the face of G×E could improve the accuracy of predicting cultivar performance in specific environments and mega-environments, thereby enhancing breeding decisions for broad or local adaptation (Jarquín et al., 2014; Xavier et al., 2025).

Maize (*Zea mays* L.) presents particular challenges and opportunities for genomic prediction across diverse environments. Maize germplasm is extremely diverse and structured into heterotic groups (e.g., “stiff-stalk” vs. “non-stiff-stalk” lineages), with clear genetic differentiation between them (White et al., 2020). Maize population structure can affect prediction accuracy if not accounted for, as genomic relationships are stronger within groups than between groups (Romay et al., 2013). Maize is also grown across diverse environments. Incorporating nonlinear, kernel-based relationship models can capture complex genetic similarities and even account for cryptic population structure in prediction models (Gianola & Van Kaam, 2008; Hu et al., 2023). Kernel methods have been applied in maize and other crops to model epistatic or non-linear effects, sometimes improving predictive power for complex traits (Hu et al., 2023).

In practice, breeding programs introduce and develop new germplasm over time as older germplasm is recycled through recombination and migration into new germplasm. Older germplasm may not predict newer target population of genotypes (TPG) well due to different haplotypes, lower genome-wide relationships, and novel linkage disequilibrium patterns (Habier et. al 2013). This “temporal cohort” dynamic means that genomic prediction models trained on past data face difficulties predicting performance of future, genetically differentiated entries. Moreover, year-to-year weather variability creates new environmental conditions not previously observed in training data. Genomic prediction accuracy drops when the target population of environment (TPE) differs from the multi-environment trials (MET) that feed the genomic prediction models (Rogers & Holland, 2022). Rogers and Holland (2022) found that incorporating G×E terms improved prediction of maize yield only when the test environment was similar to those used in training, whereas predictions of novel environments remained poor.Therefore, models must capture specific environment-genotype combinations which, in turn, require large training sets with sufficient environmental and genetic variation for the models to "learn" generalizable main effect (G, E) and interaction (GxE) patterns.

The Genomes to Fields (G2F) initiative began in 2014 to leverage collaborations across public institutions growing unified sets of germplasm across numerous environmentally distinct locations and years (environments). Replicating a unified set of hybrids at 20-30 locations each year, phenotypic variability can be more accurately partitioned into estimates of genotype, environment, and G×E effects. Changing the germplasm every two years (with a small proportion of entries represented by long-term check hybrids), G2F has increased the genetic breadth of hybrids evaluated. Prior to this competition, G2F planted and harvested experiments that cumulatively included more than 6,000 distinct hybrids and 295 distinct location-year environments. Phenotypic and genotypic data for these field trials were collected, curated and made publicly available in a standardized format facilitating comparisons. G2F is the best available public dataset for studying genomic prediction model development and effectiveness.

A previous version of the G2F dataset was used as the basis for a phenotype prediction competition in 2022 and 2023 (Washburn et al., 2025, Lima et al., 2023). That competition attracted 241 participants, resulting in development of numerous publicly available models, scientific collaborations, and the training of young scientists from across the United States and globally. The G2F community determined that future competitions, making additional past years available (2014-2023) while hiding the current year (2024), would be valuable to advance scientific and practical understanding.

Here we describe the public maize yield prediction competition held from November 15, 2024 to January 15, 2025. Participants were provided with a curated version of the openly available G2F G×E project dataset containing ten years (2014 to 2023) of maize yield trial data, sequencing-based genotypes, and weather data from across the United States to develop predictive models (Chen et al., 2026). Participants submitted their model's grain yield predictions for the 2024 growing season. The G2F data from 2024 were made publicly available only after the competition.

Materials and Methods

Data Source

Public maize hybrid yield data from the G2F initiative 2014-2023 was previously made publicly available (Chen et al., 2026; Lima, et al., 2023, 2023, 2023; McFarland et al., 2020). Data were curated, standardized, formatted, and filtered following the methods used in the previous competition as described in Washburn et al. 2025. For this competition, an extensive description of the dataset, division between training and testing data, and other relevant details can be found in Supplemental File S1. A significant change from the previous competition was the size of genotypic data. Following feedback on the 2022/2023 competition that the genotypic data were inaccessibly large for some participants, we filtered the genotypic data to 2,425 variant positions with the aim of increasing participation. (Supplemental File S1). As in the previous competition, weather data was downloaded from NASA Power (https://power.larc.nasa.gov/) and environmental covariate data was generated using a previously described pipeline (Lopez-Cruz et al., 2023; Washburn et al., 2025). Although weather data from field weather stations were available for all sites, variations in station installation, and maintenance likely contributed to increase noisy in the data. As a result, we relied on NASA weather data, which provided a more standardized and uniform source of information across locations. While these data may not fully reflect site-specific microclimatic conditions, their consistency made them preferable for comparative analyses in the competition. . The data used (including the test set which was not provided to participants until after the competition) covered eleven years and numerous locations in the United States, and Canada (Figure 1). The combined dataset comprised 295 unique environments (i.e., locations-years combinations) and more than 6,000 unique hybrids. The final dataset is publicly available at https://www.genomes2fields.org/resources/ and on CyVerse (https://doi.org/10.25739/78mn-4394) (Chen et al., 2026).

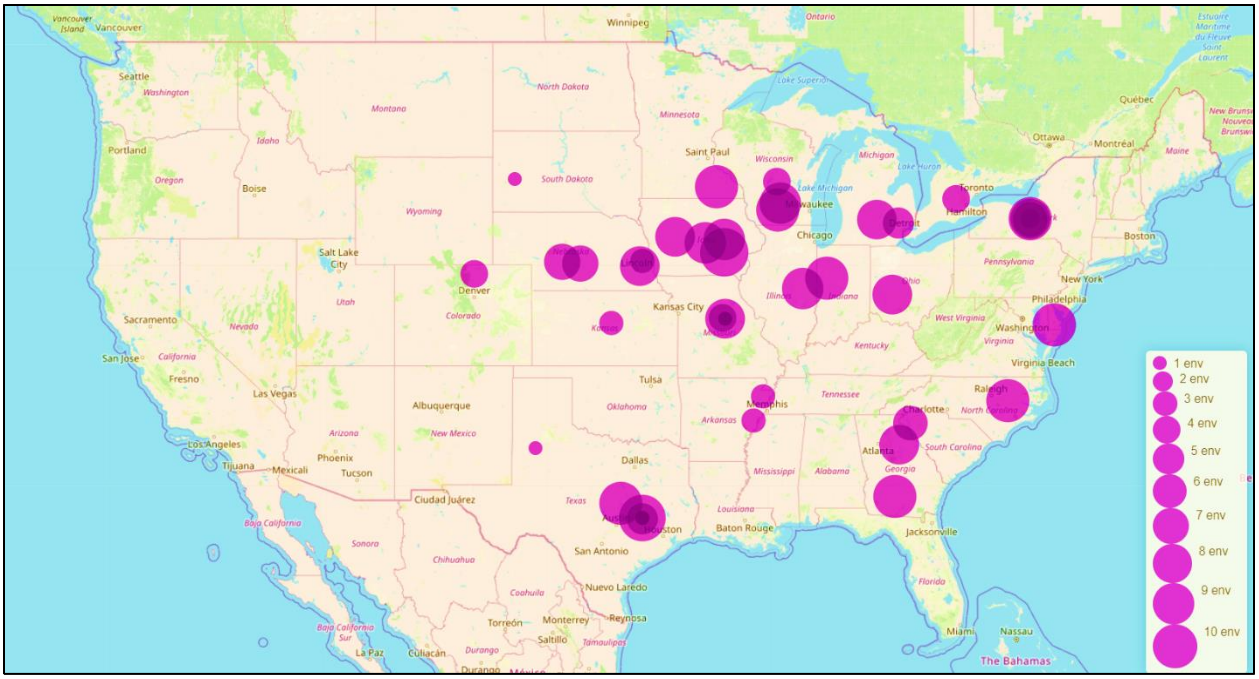

Figure 1. Geographic Distribution of Experimental Environments Across the United States. Environmental sites tested in the United States and Canada from 2014 to 2024. Map showing the spatial distribution of experimental environments included in the study. Each circle represents a location where experiments were conducted, and circle size indicates the number of environments evaluated at that location in different years.

Competition Methods and Rules

The competition was run as previously described Washburn et al. (2025) with limited modifications. Complete instructions, methods, and rules for the competition as presented to the participants are in Supplemental File S2. The EvalAI platform (Yadav et al., 2019) was used to host the competition leaderboard, submissions, and evaluation scripts. EvalAI was the primary method by which participants interacted with the competition. As participants submitted their predictions to the platform, each was evaluated in real time without providing the test set data to the participants. Resulting prediction accuracy was displayed publicly on the leaderboard. The former competition's winning approach was entered as the benchmark method.

After the previous (first) competition, it was determined that the community and many participants would prefer to use Pearson's correlation coefficient (r) instead of Root Mean Squared Error (RMSE) as the metric in order to place greater emphasis on the relative rankings of hybrids than on the absolute predictions. Therefore, the model evaluation metric used here was the average within-environment Pearson's r, measured as the average value of r between predicted and observed values estimated *within* environments of the test set. Calculating the metric within each environment first and then averaging that value, rather than calculating across the whole data set, was considered critical in both this and the previous competition in order to emphasize performance at each location.

Individual Team Methods

Teams were voluntary, global, and discovered the competition through various channels [email, advertising to X and Y listserv, public website, etc....]. Each team employed their own knowledge and methods without guidance from G2F program staff. To reduce lucky guesses, overfit models, and "cheating", each team was limited to three submissions without any feedback outside of the public leaderboard. Nevertheless, it was discovered that some teams unethically developed multiple accounts and shared results. Only the final of the three attempts for each team is reported.

Teams in the competition employed various methods and strategies. Common methods included best linear unbiased prediction (BLUP), random forest, gradient boosting, neural networks, and support vector regression. Many teams also used combinations of models in ensembles. Full details of each team's methods, for those XX out of YY teams that shared them, can be found in Supplemental File S3.

Winning Team Methods

The winning model employed a multivariate mixed linear model that parameterized genetics using a Gaussian kernel. This kernel meant to capture non-linear relationship among individuals, which were hypothesized to account for major population structure and genotype-specific adaptation. Each location-year combination was treated as a distinct response variable to allow for environment-specific genetic effects. The Gaussian kernel was constructed as

$$\mathbf{K} = \exp(-\mathrm{h}\mathbf{D}^{\mathbf{2}}),$$

where $\mathbf{D}$ was the Euclidean distance matrix among genotypes, built from molecular markers, and $\mathbf{h}$ was the bandwidth parameter calculated as the inverse mean value of $\mathbf{D}$.

The multivariate model was fit using the "ZSEMF" function from the R package bWGR (Xavier et al.,2019), corresponding to the MegaSEM approach (Xavier and Habier 2022, Xavier et al., 2025). The model for the environment *k* is described as

$$\mathbf{y}_k = \boldsymbol{\mu}_k + \mathbf{g}_k + \mathbf{e}_k,$$

assuming unstructured covariance among environments described by

$$\mathbf{g} \sim \mathrm{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathrm{g}} \otimes \mathbf{K}),$$

where $\boldsymbol{\mu}_k$ is the overall mean of the environment k, $\mathbf{g}_k$ is a vector of genomic values, $\mathbf{e}_k$ is the vector of residuals for the environment k, and the $\boldsymbol{\Sigma}_{\mathrm{g}}$ describes the variance-covariance matrix of genotype values across environments. Predictions of the unobserved year were computed as the average prediction from previous years from the corresponding locations, assuming those would correspond to the best samples of the target population of environments (TPE). All computations were performed in R (R Core Team 2024) with use of the "bWGR" package (Xavier et al.,2019).

Results and Discussion

Participation

A total of 370 individuals within 205 teams registered for the competition. These registrants came from 42 countries with around half of the registrants coming from within the United States. Most teams consisted of only one member with the largest team being made up of nine. Most participants came from academic institutions (67%) followed by private individuals (10%), and private companies (10%). Self-reported government and non-profits comprised the remainder of the participants with just under 7% coming from each of those categories.

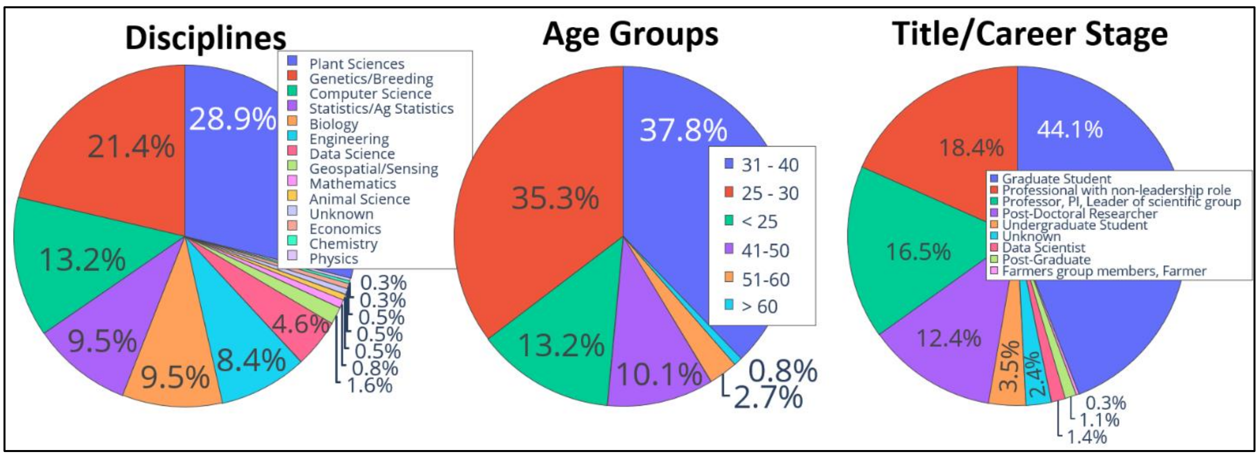

Figure 2. Demographic and professional profile of competition registrants. Distribution of competition registrants by academic discipline, age group, and title or career stage.

Competition registrants came from a large variety of disciplines. The largest group of registrants (nearly 30%) were from Plant Science disciplines (Figure 2). The second largest group (21%) came from Genetics or Breeding, followed by Computer Science (13%), Statistics (10%), and Biology (10%) (Figure 2). The remainder came from various disciplines including Engineering, data science, mathematics, animal science, economics, physics, chemistry, among others (Figure 2). A plurality of registrants (44%) were graduate students, followed by professionals in non-leadership roles (18%) (Figure 2). There was one individual who selfidentified as being a farmer, a group that stands to gain from improved predictive models but is often not heavily involved in designing them. Registrants were predominantly under the age of 41 with the largest segment (38%) in the age range of 31-40 (Figure 2). The second largest segment was ages 25-30 (35%), followed by individuals who were less than 25 years old (Figure 2).

Prediction Rankings

The final team rankings for the top ten fully qualified teams appear in Table 1. Although not shown, the benchmark model (winner of the previous edition) slightly outperformed the winning model from the current competition. Some high-ranking teams also withdrew or were disqualified and not included in the final rankings. Figure 3 shows the overall distribution of scores from all qualified submissions to the competition. The distribution is consistent, with many submissions scoring in the 0.30 to 0.40 range but only a few achieving the most competitive high scores above 0.40. These prediction scores are in line with what is typically seen for yield prediction from large real-world datasets like this one. However, the volume of environments tested was ~ 10X what has previously been reported, outside of the previous competition. (Washburn et al. 2025).

Table 1. Top 10 Teams Ranked by Prediction Accuracy Based on Pearson Correlation

Rank	Team Name	Pearson (r)
1	PARaBra	0.437
2	The Cornquerors	0.426
3	GxE4GoodY	0.414
4	LisbonBio	0.414
5	G2Amours(G_et_E)	0.405
6	UFEED	0.403
7	ORNL_CompBio	0.383

8	MPB_Group	0.379
9	MAMAMYA	0.371
10	MmdlForGxe	0.370

Ranking of the top 10 participating teams based on Pearson correlation coefficient (*r*) between predicted and observed values. Higher Pearson *r* values indicate stronger predictive performance.

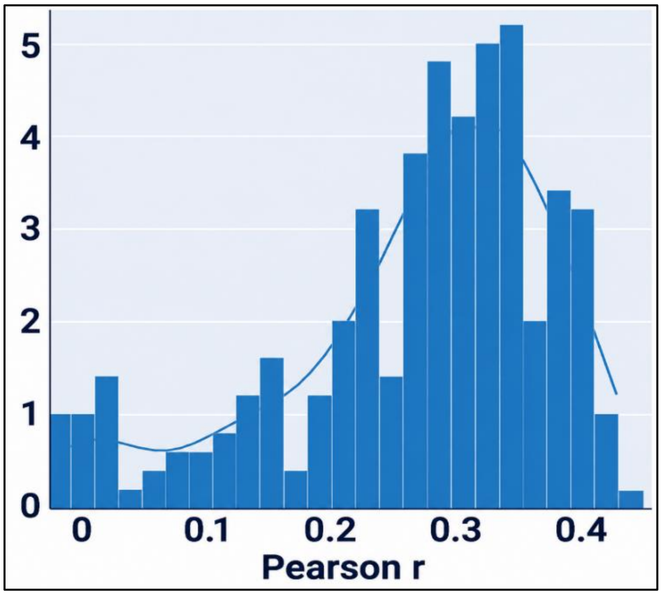

Figure 3. Distribution of prediction accuracies across models. Distribution of prediction accuracies measured as average Pearson correlation coefficients across environments.

Modeling Strategies

A wide diversity of strategies was used in the competition showcasing the many common modeling methods in use today as well as the essentially limitless number of combinations, tactics, and strategies (Figure 4A, Supplemental Table S1). Categorizing methods is challenging as there was/ could be significant overlap in categories and subjectivity involved. As was the case in the previous competition, ensembles that combined multiple model types together for a final prediction were used extensively. BLUP models were also used by many of the teams. Random forest models were also commonly used, as well as gradient boosting with XGBoost and other methods. Deep learning, neural network, and transformer methods were used by 10%, 7%, and 7% of the teams respectively, though some of these categories might have overlapped. While 24% of the teams used methods here grouped into the “other” category, other methods were diverse and used by only one or a few teams.

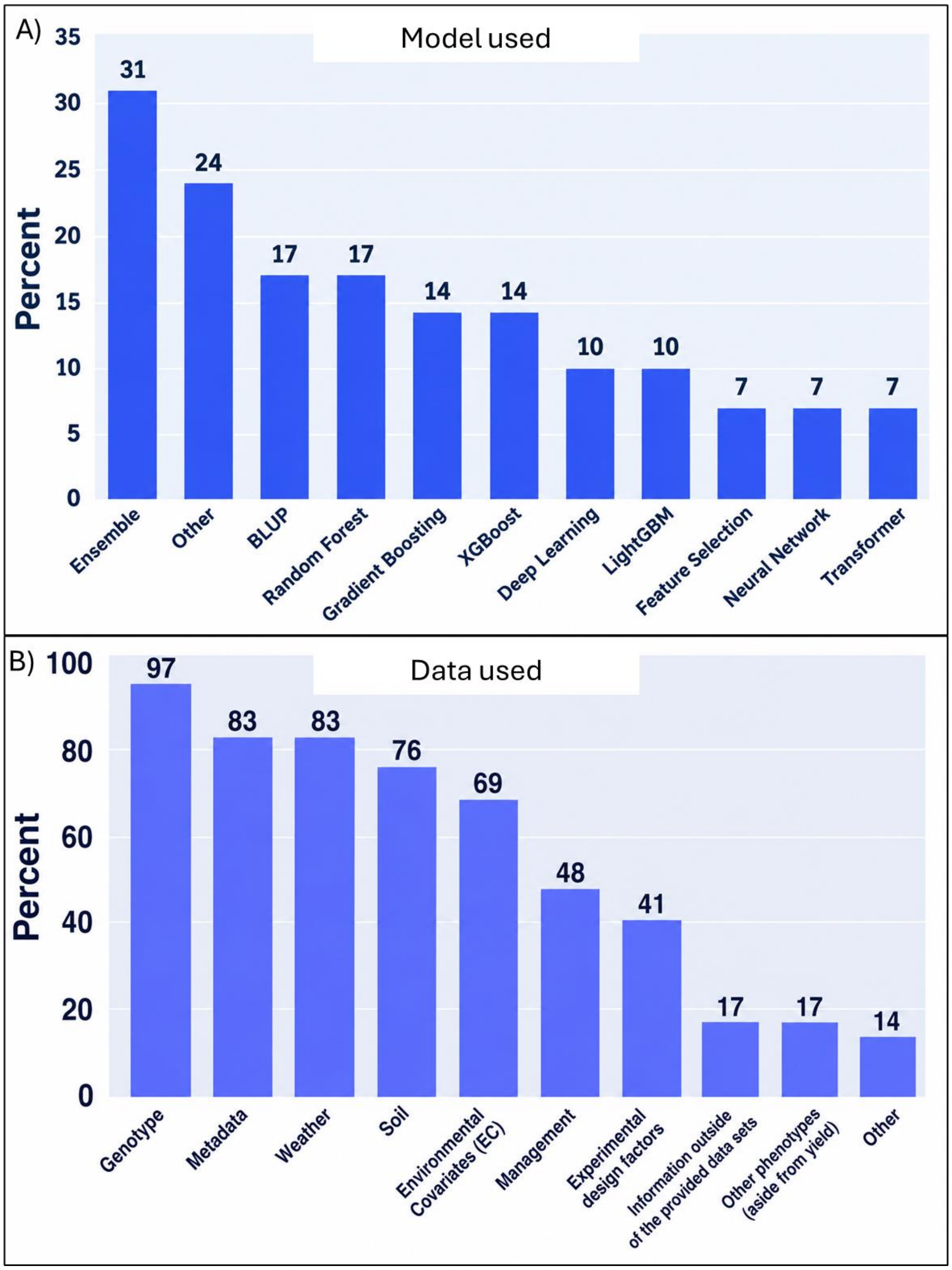

Figure 4. Modeling factors and methods used by participating teams. Panel A: Frequency of modeling approaches used for prediction. Percentage of studies or analyses using different prediction approaches. **Panel B: Frequency of predictor data types used in prediction models.** Percentage of studies or analyses incorporating different predictor data sources.

The teams also used a variety of model input factors (Figure 4B). 97% of the teams used genetics in their model. Most teams also used the provided field metadata, weather, soil, and environmental covariate data. Less than half used the provided management, experimental design information, or the agronomic phenotypes (besides grain yield). A few teams used other data that was not provided by the competition and was publicly available from other sources.

Winning Team Results

The multivariate Gaussian kernel model used by the winning team was the most successful model at predicting the grain yield of new hybrids in new environments. Across the 22 distinct environments in 2024, yield correlation (r) between predicted and observed yields ranged from 0.25 to 0.60, with an average of ~0.43 (Figure 5). This level of accuracy is encouraging given the difficulty of the scenario. It is comparable to the predictive abilities reported in other maize studies for untested hybrids in a new year. For instance, a recent multi-environment genomic prediction study in maize reported prediction correlations of ~0.34 for across-year scenarios with previously tested hybrids and ~0.21 when predicting entirely new hybrids (Barreto et al., 2024). However, the specific population evaluated in 2024/2025 included presumably greater genetic diversity, since the G2F entries are not specifically what were selected by a breeding program; while also including a much greater diversity of environments. The winning average (~0.43) likely reflects the benefit of the Gaussian kernel capturing population structure and possibly non-linear effects, as well as the inclusion of many years of training data which can stabilize predictions across environments. Notably, the spread in accuracy indicates that prediction performance varied considerably by environment (Figure 5). This outcome underscores the limitation of the model’s environment-independence assumption - without explicit environmental covariates, the model relied on genetic relatedness and the average responses learned from past environments. Many of these past environments were different years at the same locations, which could have different weather but similar soil and agronomic practices. Therefore, extreme or unique environmental scenarios in 2024 might have posed a challenge, suggesting an avenue for improvement by integrating environment descriptors or G×E terms in the future. The model's yield predictions for 2024 new hybrids also provide insight into how well genomic selection might work in practice. For breeders, this means that selecting top candidates based solely on genomic predictions could enrich good performers but would not be perfectly reliable. In a real breeding population, the results would be expected to be lower because many poorly performing inbreds and hybrids made from within the same heterotic groups would have been discarded prior to multi-environment testing. In 2024 G2F dataset, the inclusion of such hybrids (for various purposes) would have inflated overall prediction correlations. However, real breeding populations could also show actual improvement over this G2F dataset because 1) the target population of breeding program environments is more targeted (e.g. G2F locations ranged from Canada to Texas, while breeding programs typically focus on only one of the 5? maturity zones within this range); and 2) breeding programs would typically train on a single population of germplasm and recycle this germplasm for testing and validation, whereas 90% (?) of the training populations of G2F (2015-2024) were pulled from separate diverse populations. Therefore, this G2F competition dataset gets at the average genetic predictability of elite U.S. maize rather than the predictability of a specific population for a specific target environment.

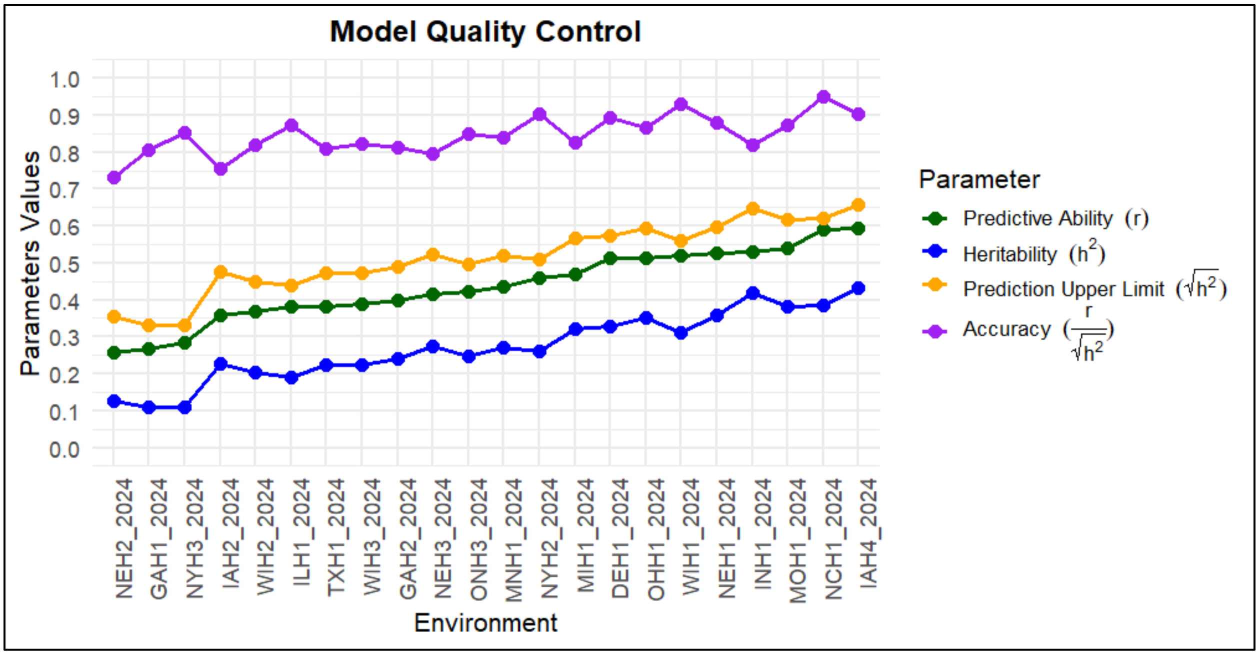

Figure 5 Model Quality Control. On the x-axes represent each new environment predicted on the 2024 data set by the winning model. On the y-axes, parameter values ranging from 0-1. Each parameter is represented by a color. Within the chart, the first line informs Accuracy, the second Prediction Upper Limit, the third Predictive Ability, and the Fourth Heritability

Impact of Kernel-Based Modeling

The use of a Gaussian kernel in the winning model was motivated by the diverse genetic background of the hybrids. Theoretically, the Gaussian kernel mitigated this by effectively grouping genetically similar individuals and down-weighting distant relationships. This can be interpreted as the kernel accounting for population structure. The Gaussian relationship matrix does not force a constant relationship for all pairs of hybrids; instead, it can allow each subpopulation to form its own cluster of higher relatedness (Figure 6). These results align with findings in wheat by Hu et al. (2023), who reported that using weighted Gaussian kernels to reflect subpopulation genetic characteristics improved multi-environment prediction by up to 31%. While the winning implementation did not incorporate explicit GWAS-based weighting as in that study, it confirms that the kernel can matter when dealing with structured populations. In practice, breeders working with diverse germplasm could benefit from non-linear kernels or at least stratified models that capture major genetic groups, to avoid biases and make full use of data.

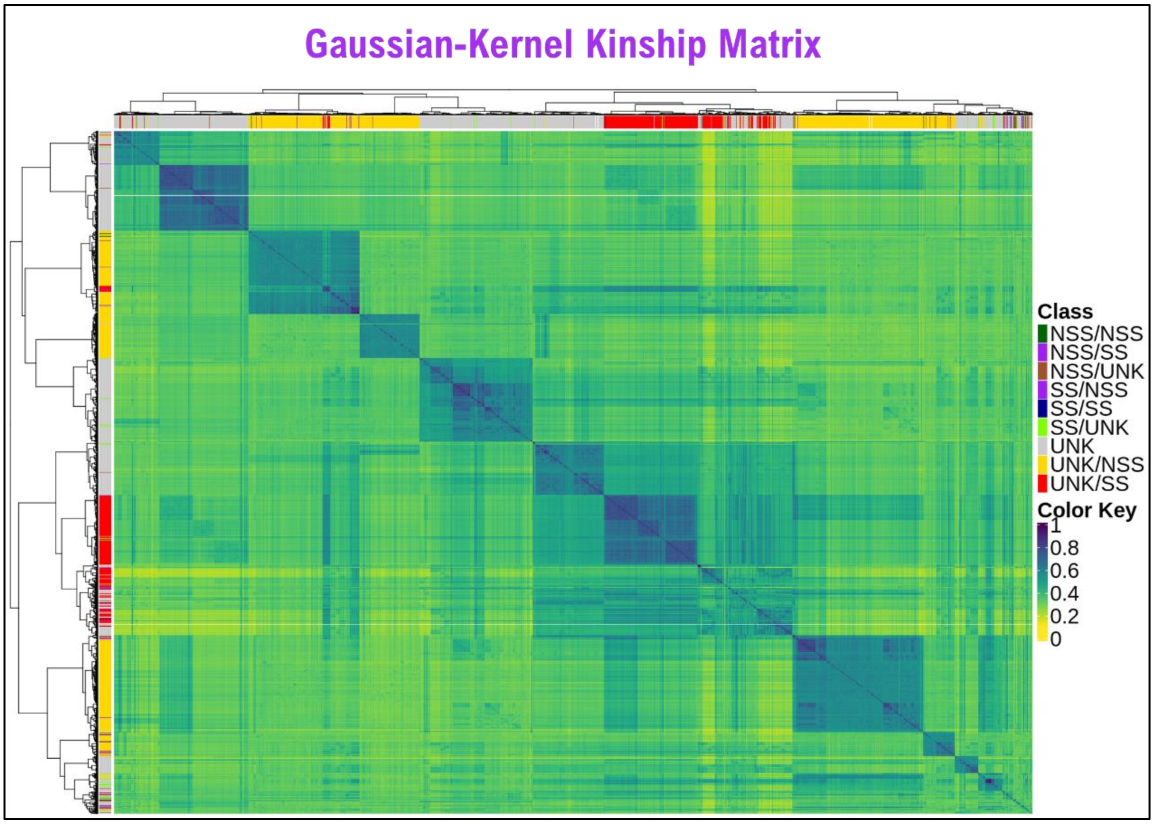

Figure 6 Gaussian Kernel Kinship Matrix. Hybrids are clustered together and its proximity is measure from 0 to 1. On the column and the top of the matrix, there is also a classification of the Hybrid parentage in "stiff-stalk" and "non-stiff-stalk" represented in the legend by different colors.

Incorporating Genotype × Environment Interactions

The winning approach accounts for G×E by estimating the genetic correlations of yield performance among all pairs of environments from the unstructured covariance matrix (Figure 7). No environmental covariates were utilized, thus the G×E patterns captured by the winning model were not limited to those that can be captured by environmental factors. When the target environment was unprecedented (low similarity to training), the accuracy suffered. Future work can leverage that resource to refine the present analysis- e.g., by using crop-model simulated features like drought indices or degree-days to explain G×E. Another notable effort by Costa-Neto et al. (2021) introduced an "enviromic assembly" approach, wherein key environmental typologies are identified and used to derive environmental similarity kernels. Costa-Neto et al. (2021) tested this in tropical maize and improved the prediction of yield plasticity, even allowing for “virtual” evaluation of genotype–environment combinations to optimize field testing. This suggests that by intelligently summarizing environments, one can better model why certain hybrids perform well in some conditions and not others.

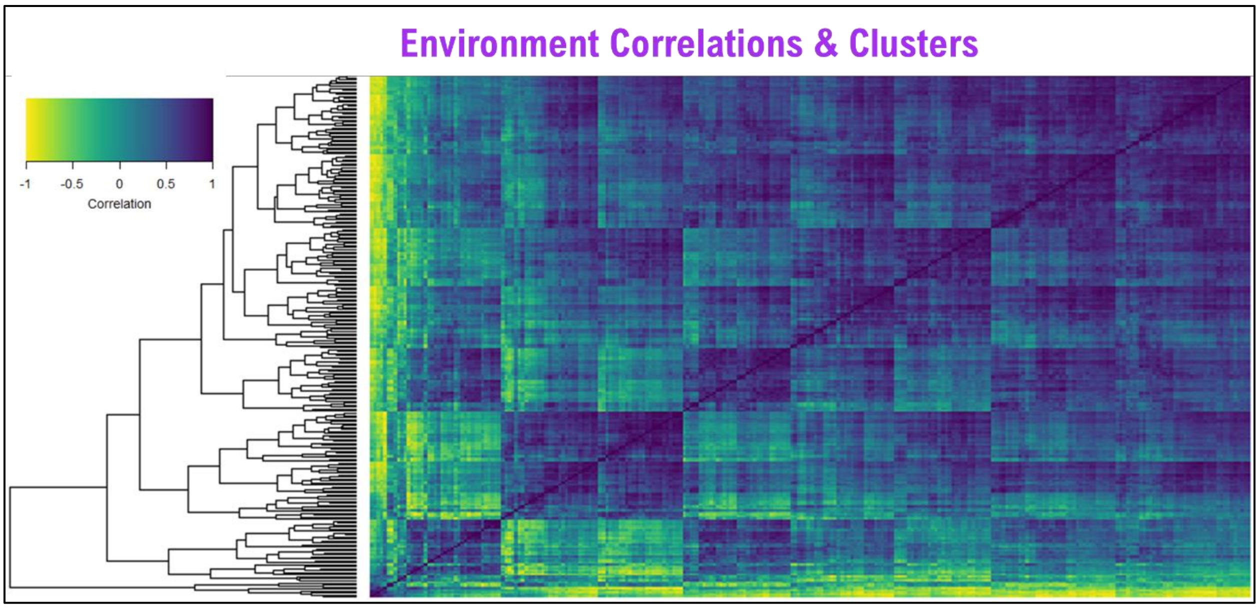

Figure 7 Genotype Environment Correlations and Clusters. On the left trees aggregate the environments with genetic similarities. Correlations range from -1 to 1 and are colored from yellow (-1) to dark blue (1).

Practical Implications for Maize Breeding

From a breeder’s perspective, these findings reinforce both the promise and the limitations of genomic prediction across mega-environments and the ability to train with different germplasms than was tested. On the positive side, the model showed that even without explicit environmental inputs, a genomic model can capture a large share of the predictable genetic variation, yielding useful predictions of hybrid performance across a broad geography and multiple years. This can facilitate early discarding of poor candidates and focus resources on promising lines, thereby increasing breeding efficiency. Importantly, the model did particularly well in environments where genetic variance was high and in predicting broadly adapted genotypes that perform consistently. This suggests that genomic selection will be especially valuable for improving yield in stable, highly agronomically managemed conditions and for selecting genotypes that are not extremely specifically adapted. Indeed, many breeding programs aim for widely adapted hybrids, and this approach appears well-suited for identifying those.

On the other hand, G×E remains a significant hurdle: no single model or method emerged as a panacea for all environments. In the previous global G2F prediction competition (Washburn et al., 2025), it was observed that diverse modeling strategies (from simple linear models to advanced machine learning) achieved roughly comparable results, and no approach could perfectly predict all environment combinations. The results of this competition echo that, with even a sophisticated kernel method having blind spots. Thus, breeders should consider an ensemble of approaches and maintain field testing, especially in new or extreme environments, rather than relying purely on genomic predictions. Reaching the eventual goal of "predicting phenotype from genotype and environment" is advancing, but will likely require integrating crop growth models or deep learning that can capture nonlinear G×E, as suggested by recent work (Barreto et al., 2024). In the interim, pragmatic use of genomic selection in maize might involve multi-tier testing: use genomic predictions to eliminate or advance major outliers, deciding which environments are most informative to test, and then relying on actual multi-environment field data for final decisions among top candidates.

Post-competition ensembles outperformed the best individual models

As with the previous competition, many of the participants used ensemble methods to combine multiple disperate models into a single prediction result. Post-competition, simple averaged ensemble models were created to test if diverse models developed by different competitors had distinct and complementary predictive strengths. For simplicity, pairwise models were tested between each of the model on the final leaderboard. This resulted in over 1,000 ensemble models of which 27 had higher accuracy than the winning team. Interestingly, all of these 27 models included the first or second place team as one of the two parent models. The other parent was highly variable in its original competition ranking. The best pairwise ensemble was a combination of the winning model and the third-place model and achieved a mean Pearson r value of 0.458 as compared to the 0.437 value of the winning model (Figure 8). Interestingly, an ensemble of the first and second place models did not perform better than the first-place model alone. The errors for these two parent models are highly correlated (r = 0.927) whereas the first and third place models, the highest performing pairwise ensemble, had errors with a reverse correlation (r = -0.363). The methodologies of these models are also substantially different. Although theoretically possible, few notable examples of two poorly performing models contributing to a high accuracy ensemble were found. One notable exception was the combination of the 11^{th} and 12^{th} place models which produced an increase of nearly 0.05 in the correlation coefficient over the highest performance of the two parent models. This improvement was not enough to overtake the top individual models but is another example of diverse error values (r = 0.153 between the two model's error values) combining for higher performance.

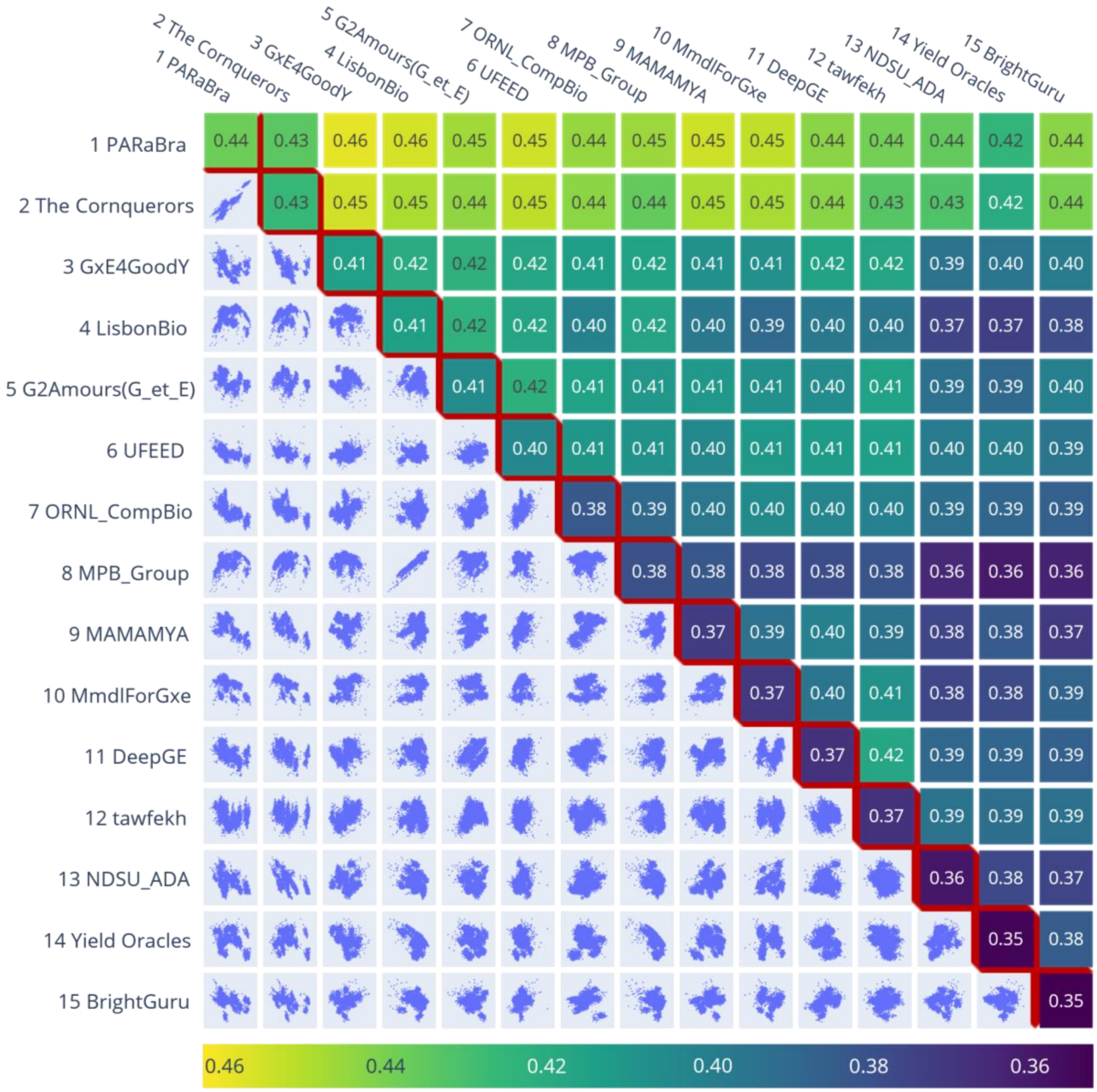

Figure 8 Pairwise ensemble models of the top 15 teams. The lower left triangle shows a scatter plot for each pairwise model's error values. The top right triangle shows the Pearson r score for each pair's ensemble. The diagonal displays the original score of each model by itself.

In conclusion, this study presents the development and results from dozens of yield prediction models tested on a single dataset within an international competition. The results underscore the conclusion that many different methods can perform well, and no single method is far and away superior to all others. The winning model demonstrated the value of non-linear, multivariate genomic prediction across a wide range of environments and years. By leveraging a decade of multi-environment trial data, despite from semi-independent genotypes and independent years, the winning model provided the most accurate predictions for a future set of hybrids. Post-competition ensemble models that included the winning and some other high scoring models were able to outperform their parent models, demonstrating the value of combining the results of models with different strengths and weaknesses. Future improvements should target how to incorporate environmental information without compromising accuracy, and more flexible G×E structures. Advancing toward fully integrated genomic–environmental prediction models will help breeders more reliably predict hybrid performance in target environments, ultimately leading to more efficient development of resilient, high-yielding maize cultivars. Furthermore, this could help geneticists and others understand the biological basis of prediction and missing heritability.

Supplement Materials

Supplemental File S1

Supplemental File S2

Supplemental File S3

Supplemental Table S1

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